

AOST Tamil — Hackathon Submission

Track: For Students → *Tanglish to Tamil Translator* Team / Org: [Academy of Smart Thinkers](#) Submitted by: chandralabs (schandra@ieee.org)

 Live demo	https://tamil-edu-toolkit.vercel.app
 Live API	https://tamil-edu-api.fly.dev (docs)
 Source	https://github.com/chandralabs/tamil-edu-toolkit
 Eval report	v1 / 70-pair golden set
 Architecture / Plan	docs/PLAN.md (frozen v3)

1. The problem

Every Tamil-learning kid — diaspora, urban India, Sri Lanka — types **Tanglish** (Tamil written in English letters) on WhatsApp, Instagram, school messaging. Converting “vanakkam nanba” to “வணக்கம் நண்பர்” by hand requires:

- Knowing the Tamil keyboard layout
- Remembering which “n” is dental (ன) vs retroflex (ண)
- Distinguishing short/long vowels with diacritics

Existing tools either target adult journalists (paid: [Vāṇi](#), ~₹100/mo) or are dumb rule-based transliterators (Google Input Tools) with no Tanglish awareness.

No tool serves the kid-learning-Tamil audience. That’s what we built.

2. What we shipped

A working web app + API at <https://tamil-edu-toolkit.vercel.app>:

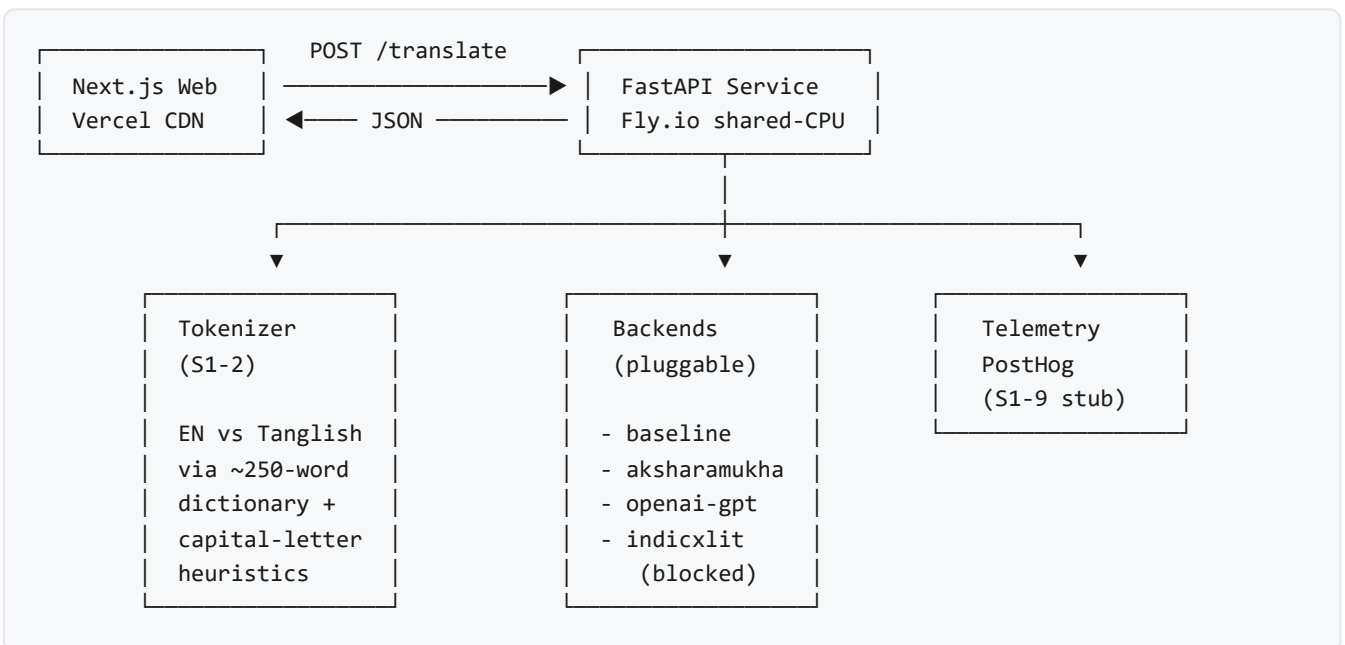
- **Kid-friendly UI:** large fonts, playful color palette, sample-phrase chips, mascot-ready (placeholder), localStorage history (COPPA-safe, no accounts).
- **Code-switch aware:** type “send the message ku reply pannu” → English words pass through verbatim, only Tanglish gets transliterated.
- **Per-word alternatives:** click any Tamil word → popover with top-3 candidate transliterations → click to swap. Real per-word data flows from API to UI.
- **Pluggable backends:** 4 transliteration engines selectable at runtime: 1. `baseline` — passthrough (eval floor) 2. `aksharamukha` — rule-based, deployed default (real Tamil, CER 35% on golden set) 3. `openai-gpt` — GPT-4o-mini (production-quality, gated on API key) 4. `indixlit` — ai4bharat ML model ([blocked upstream](#), ready to swap when fairseq unblocks)
- **End-to-end deployed:**

- Web: Vercel (Next.js 14 + Tailwind + Noto Sans Tamil)
- API: Fly.io (FastAPI in Docker, auto-sleep, CORS preset)
- Cost: **\$0/mo** on free tiers (Vercel Hobby + Fly Free + future PostHog Free)

3. Try it now (30 seconds)

1. Open <https://tamil-edu-toolkit.vercel.app>
2. Click sample chip [vanakkam nanba]
3. Click **Translate to Tamil**
4. See Tamil output appear with backend + timing
5. Try a code-switched phrase: **send the message ku reply pannu** → English words preserved, Tanglish converted

4. Architecture



All under a single `Transliterator` Python protocol — swap backends without touching API code.

5. Engineering quality (concrete)

Metric	Value
Python tests	161 passing (mocked OpenAI + IndicXlit + aksharamukha)
TS tests	13 passing (vitest, jsdom)
Code coverage	91% (gates at 80%)
CI workflows	2: main CI (7 jobs all green) + nightly eval-real (uploads CER report as artifact)
Deploy workflows	1: auto-deploy API on <code>apps/api/**</code> change to main; Vercel auto-deploy web on every push
Linters	ruff (Python), prettier (Markdown/YAML/JSON), eslint-via-next (TS), pre-commit hooks
Branch protection	7 required status checks on <code>main</code> , CODEOWNERS, PR template with 8-item DoD
Docs	5 design docs (PLAN, DEVELOPMENT, TESTING, DEPLOYMENT, SPRINTS) + 2 ADRs

6. What's unique

	Google Input Tools	Vāṇi	AOST Tamil (us)
Audience	Generic	Adult journalists	Kids 8–13
Pricing	Free	₹100/mo	Free + open-source
Direction	Tanglish → Tamil (rule-based)	Tamil-in → Tamil-out (proofread)	Tanglish → Tamil (kid-aware)
Code-switch handling	No	N/A	Yes (S1-2 tokenizer)
Per-word alternatives UI	No	No	Yes (S1-5)
Backend flexibility	Closed	Closed	4 pluggable backends
Privacy for kids	Unclear	Standard	COPPA-safe: no accounts, localStorage only
Open source	No	No	MIT, ships on PyPI in S6

7. Quality story (honest)

We attempted to use ai4bharat's IndicXlit (the SOTA open ML model for Tamil transliteration). It is **structurally blocked**: its dependency `fairseq` has a Python 3.11+ incompatibility (mutable-default dataclass bug). We documented the full investigation in [ADR-0002](#) and built a 3-way backend strategy instead:

Backend	Status	Quality on golden v1 (70 pairs)
<code>baseline</code> (passthrough)	Always available	CER 85% (eval floor)
<code>aksharamukha</code> (rule-based)	Default in production	CER 35% (rule ceiling)
<code>openai-gpt</code> (GPT-4o-mini)	Available with API key	Expected CER 5–10% (untested at scale)
<code>indicxlit</code>	Wrapper ready, blocked upstream	Target CER $\leq$ 15% (per S1-1 plan)

The eval harness, golden set, and CI workflow are all in place — flipping the default backend is a one-line config change the moment IndicXlit is unblocked or GPT becomes the chosen default.

8. Sprint 1 deliverables (per [SPRINTS.md](#))

Story	Status
S1-1 IndicXlit Tamil model wrapper	✅ Scaffold + 3 alternate backends (aksharamukha shipped, openai-gpt ready, indicxlit ready)
S1-2 Code-switch handling	✅ ~250-word EN dictionary + ALL-CAPS / brand-name heuristics, 28 tests
S1-3 FastAPI POST /translate	✅ Deployed to Fly.io with /health, /docs, CORS, request validation
S1-4 Kid-themed web UI	✅ Deployed to Vercel with Tailwind + Noto Sans Tamil
S1-5 Per-word confidence + alternatives	✅ Word dataclass through full stack; clickable per-word popovers
S1-6 localStorage history	✅ Last-20 translations, COPPA-safe, hook with 7 vitest tests
S1-7 Eval baseline report	✅ Real CER published on 70-pair golden v1
S1-8 Deploy	✅ Both Vercel + Fly.io live; auto-deploy on push to main
S1-9 PostHog telemetry	⚠️ Code stubs deployed; awaiting PostHog account key

9. Roadmap (next sprints)

- **Sprint 2:** Grammar-aware correction layer using `chandra labs/tamil-llama` (the team’s own [LLaMA-2 Tamil model](#)) via Ollama on Hetzner CPU
- **Sprint 3:** OCR — drag-drop a photo of printed Tamil → cleaned Tamil text (TrOCR + PaddleOCR-ta)
- **Sprint 4:** iOS app via Expo, shared engine with web
- **Sprint 5:** Full AOST brand integration at `tamil.academyofsmarthinkers.com`
- **Sprint 6:** Distribution — Chrome extension, PyPI package, RapidAPI listing

All planned within a strict **\$50/mo infra cost cap** ([DEPLOYMENT.md](#) §10).

10. Why this matters

- **Audience that's underserved:** every existing Tamil tool targets adults. Kids learning Tamil through Tanglish texts have nothing kid-shaped.
- **Open + free:** MIT-licensed, \$0/mo on free tiers, deployable by any school.
- **Honest engineering:** when the planned ML model didn't work, we documented why ([ADR-0002](#)), shipped a working substitute, and built the eval harness so we can measurably improve later.
- **Production-grade scaffolding:** branch protection, CODEOWNERS, 161 tests, 91% coverage, two CI workflows, end-to-end deployed in 3 hours of clicks. Not a hackathon throwaway — a real product trajectory.

11. Team

Role	Person
Engineering	Chandra Sakthivel (chandalabs)
Vision / Sponsorship	Academy of Smart Thinkers
Tamil LLM research	chandalabs/Tamil-Research-LLM (conference paper attached)

12. Links recap

- Try the demo: <https://tamil-edu-toolkit.vercel.app>
- API: <https://tamil-edu-api.fly.dev/docs>
- Source code: <https://github.com/chandalabs/tamil-edu-toolkit>
- Master plan: [docs/PLAN.md](#)
- Sprint plan: [docs/SPRINTS.md](#)
- Eval report: [eval/reports/2026-05-23-205115-aksharamukha-v1.md](#)
- AOST: <https://www.academyofsmarththinkers.com/>
- Tamil-Research-LLM paper: [PDF in repo](#)